

FINAL EXPLANATION: ESSENTIAL MATHEMATICS GRADE 10

Teacher Key: Exemplar Answers and Marking Guide

FINAL EXPLANATION: ESSENTIAL MATHEMATICS GRADE 10: 2.3 TRIGONOMETRY

Teacher Key: Exemplar Answers (do not give to students)

INSTRUCTIONS GIVEN TO STUDENTS

This is your Final Explanation. It answers the driving question: How can we use the angles and one side of a right-angled triangle to find any unknown height or distance in our environment, without ever measuring it directly? It also answers the key inquiry question: Why is trigonometry important? Write your answers in the space under each prompt. Use what you observed and learned in Lessons 1 to 8 and the Kilimani Secondary School water tank in Nakuru as your main example. Use a calculator in DEG mode. Give trig values to 4 decimal places and final lengths to 1 decimal place with units. Show every step of your working, draw labelled diagrams where asked, and use the words hypotenuse, opposite, adjacent, sin, cos and tan carefully.

Part 1: The triangle and the ratio that makes the method work

At Kilimani Secondary School nobody can climb the steel-framed water tank tower, but Wanjiru and Otieno can still find its height. (a) Draw and label the right-angled triangle formed by the tower, the level ground and the line of sight for Wanjiru, who stands 8 m from the base. Mark the angle θ at her eye, and label the hypotenuse, opposite and adjacent sides. Say what each side represents in the real situation. (b) Using the three 35° triangles A, B and C from Lesson 2 (sides 4.0, 2.8, 4.9 cm; 8.0, 5.6, 9.8 cm; 12.0, 8.4, 14.7 cm), explain what you found about the ratios when the triangles got bigger. (c) Explain why this idea matters for the tank.

- (a) The diagram is a right-angled triangle. Its horizontal side runs from Wanjiru's eye, level with her eye height, to the tower. Its vertical side is the part of the tower above her eye level. The right angle is where the horizontal line from her eye meets the tower. The line of sight from her eye to the top of the tank is the hypotenuse, which is opposite the right angle and is the longest side. The angle θ is at her eye, between the horizontal and the line of sight. The opposite side is the part of the tower above her eye level, which is across from θ . The adjacent side is the 8 m horizontal distance from her to the tower, next to θ and not the hypotenuse.
- (b) Triangles A, B and C all have a 35° angle but different sizes. In every one, $\text{opposite} \div \text{hypotenuse}$ was about 0.571, $\text{adjacent} \div \text{hypotenuse}$ was about 0.816 and $\text{opposite} \div \text{adjacent}$ was about 0.700. These are $\sin 35^\circ$, $\cos 35^\circ$ and $\tan 35^\circ$, and the calculator gives 0.5736, 0.8192 and 0.7002, very close to our measurements. The ratios stayed the same even though the triangles got bigger, apart from small measuring differences of about 0.02.
- (c) For a fixed angle, the ratio of the sides depends only on the angle, not on the size of the triangle. So Wanjiru's small triangle and Otieno's bigger triangle at different distances can give the same height. Knowing the angle and one side lets us find another side, so we do not need to measure the height directly.

Part 2: Choosing a ratio and calculating

(a) Explain how you decide whether to use sin, cos or tan when you need to find an unknown side. (b) Use the 3-step method (label, choose the ratio, solve with units) to find: (i) the length of a Kisumu roof rafter when the horizontal span from the wall to the centre is 3.6 m and the rafter makes 30° with the horizontal; (ii) the width of a Kericho stream, when a learner on the bank directly opposite a tree walks 20 m along the bank and the line of sight to the tree then makes 40° with the bank. (c) A Watamu diver's weighted line is 12 m long and makes 50° with the horizontal surface. Find the depth of the weight.	(a) I label the triangle from the angle θ and identify the side I know and the side I want. If they are opposite and adjacent I use tan. If they are opposite and hypotenuse I use sin. If they are adjacent and hypotenuse I use cos. Then I write the equation, solve it and give the answer with units. (b)(i) The known side is the adjacent side (3.6 m) and the wanted side is the hypotenuse (the rafter), so I use cos. $\cos 30^\circ = 3.6 \div \text{rafter}$, so $\text{rafter} = 3.6 \div \cos 30^\circ = 3.6 \div 0.8660 = 4.2$ m. (ii) The known side is the 20 m walked along the bank (adjacent) and the wanted side is the stream width (opposite), so I use tan. $\text{Width} = 20 \times \tan 40^\circ = 20 \times 0.8391 = 16.8$ m. (c) The known side is the line, 12 m, which is the hypotenuse, and the wanted side is the depth, which is opposite the 50° angle. So I use sin. $\text{Depth} = 12 \times \sin 50^\circ = 12 \times 0.7660 = 9.2$ m.

Part 3: Solving the water tank problem	
Wanjiru stands 8 m from the base of the tower and reads an angle of elevation of 37°. Otieno stands 10 m from the base and reads 31°. Every learner's eye height is 1.5 m. (a) Explain how the clinometer gives the angle of elevation, and what the string does when the straw is level. (b) Write a complete solution for Wanjiru: the diagram description, the ratio and why you chose it, the calculation and the answer with units. (c) Do the same for Otieno. (d) Wanjiru is closer, so whose angle is larger? Explain why both learners should still get the same height.	(a) The clinometer is a cardboard protractor with a straw along the straight edge and a string with a washer hanging from the centre mark. When the straw is level the string hangs at the 90° mark. When the straw is tilted up to sight the top of the tank, the string moves away from 90°. The angle of elevation is the number of degrees between the string and the 90° mark. (b) The tower, the level ground and the line of sight form a right-angled triangle. The known side is the 8 m horizontal distance (adjacent) and the wanted side is the height above eye level (opposite), so I use tan. $\text{Height above eye level} = 8 \times \tan 37^\circ = 8 \times 0.7536 = 6.03$ m. Adding the 1.5 m eye height: $6.03 + 1.5 = 7.53$ m, which is 7.5 m to 1 d.p. (c) The same triangle shape, with adjacent = 10 m and $\theta = 31^\circ$, so I use tan. $\text{Height above eye level} = 10 \times \tan 31^\circ = 10 \times 0.6009 = 6.01$ m. Adding 1.5 m: $6.01 + 1.5 = 7.51$ m, which is 7.5 m. (d) Wanjiru is closer (8 m), so she has the larger angle (37° compared with 31°). Both get 7.5 m because tan is the ratio opposite $\div$ adjacent, which depends only on the angle, so $\text{height above eye level} = \text{distance} \times \tan(\text{angle})$ for each learner. Both learners are measuring the same tower, so each pair of distance and angle gives the same height above eye level, 6.0 m, and the same total height.

Part 4: Checking the answer and sources of error

Group Amani used their clinometers on the school flagpole, standing 15 m from the base and reading 27°. Mr. Kamau, the school caretaker, used a tape to check the heights: the flagpole is 9.0 m and the top of the water tank is 7.5 m. (a) Calculate Group Amani's flagpole height. (b) Work out their error. (c) Explain why small differences like this happen. (d) Use your answers from Part 3 and Mr. Kamau's tape value for the tank to judge whether Wanjiru's and Otieno's answers are good.

- (a) The known side is the 15 m horizontal distance (adjacent) and the wanted side is the height above eye level (opposite), so I use tan. Height = $15 \times \tan 27^\circ + 1.5 = 15 \times 0.5095 + 1.5 = 7.64 + 1.5 = 9.14$ m, which is 9.1 m to 1 d.p.
- (b) The error is $9.14 - 9.0 = 0.14$ m, about 0.1 m when the heights are given to 1 d.p. ($9.1 - 9.0$).
- (c) The angle is read from the clinometer to the nearest degree, so it can be up to half a degree out. Small differences in where the string rests, the straw not being held perfectly steady, and distances measured along the ground also add small errors. Because the angle is rounded, the calculated height is close to the real height but not exact.
- (d) Mr. Kamau's tape value for the top of the tank is 7.5 m. Wanjiru's answer (7.53 m) and Otieno's answer (7.51 m) both round to 7.5 m, so they agree with the tape within the rounding error from reading angles to the nearest degree. This shows the method works.

Part 5: Answering the driving question

Write a clear explanation in your own words that answers: How can we use the angles and one side of a right-angled triangle to find an unknown height or distance in our environment, without measuring it directly? Use the Kilimani Secondary School water tank as your example. Include: the triangle, why the method works, the steps, and how you know the answer can be trusted. Then answer the key inquiry question: Why is trigonometry important? Give at least two real-life Kenyan examples.

We can find the height of the tank without climbing it because the tower, the level ground and the line of sight form a right-angled triangle. We measure the horizontal distance along level ground with a tape and the angle of elevation with a clinometer. For a given angle, the ratio opposite ÷ adjacent (tan) is the same whatever the size of the triangle. That is why tan of the angle links the distance to the height above eye level: height above eye level = distance $\times$ tan(angle). For Wanjiru, $8 \times \tan 37^\circ = 6.03$ m, and adding the 1.5 m eye height gives 7.5 m. For Otieno, $10 \times \tan 31^\circ = 6.01$ m, and adding 1.5 m also gives 7.5 m. They stand at different distances and read different angles, but they agree because both are measuring the same tower and the ratio depends only on the angle. To solve any problem I label the triangle, choose sin, cos or tan from the known and wanted sides, calculate in DEG mode, add the eye height when needed, and state the answer with units. I can trust the answer because both learners agree and because Mr. Kamau's tape check of 7.5 m matches within the small error from reading angles to the nearest degree, as with the flagpole where we got 9.1 m against 9.0 m.

Trigonometry is important because it lets us find heights and distances that are too difficult, too dangerous or impossible to measure directly. Surveyors use it to map the Kenyan highlands, engineers use it to design the Thika Superhighway overpasses, and foresters use it to measure tree heights in the Mau Forest without felling the trees. It is also used by a Rift Valley farmer to estimate the height of a cliff, or by builders to find the length of a Kisumu roof rafter.

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Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
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Right-angled triangle model and side naming	Draws and labels the triangle correctly from θ , identifies the hypotenuse, opposite and adjacent sides, and links each to a real part of the tank situation.	Labels the triangle and sides mostly correctly, with a weak link to the real situation.	Labels are missing or incorrect, or opposite and adjacent are confused.
Understanding why ratios stay constant	Clearly explains that sin, cos and tan depend only on the angle, using the 35° triangle evidence, and links this to Wanjiru and Otieno getting the same height.	States that the ratios stay the same but gives limited evidence or a limited link to the tank.	Does not explain why the ratios stay constant, or says they change with triangle size.
Choosing the ratio and calculating	Chooses the correct ratio from the known and wanted sides with a reason, calculates accurately with the 1.5 m eye height where needed, and gives correct units and rounding.	Chooses the correct ratio and calculates mostly correctly, with a small slip in units, rounding or eye height.	Chooses the wrong ratio or makes major calculation errors.
Checking results and explaining error	Compares calculated heights with the tape values, finds the small error of about 0.1 m and explains it by reading angles to the nearest degree.	Compares with the tape values but gives a limited explanation of the error.	Does not compare with the tape values or gives no valid reason for the error.
Answering the driving and key inquiry questions	Gives a complete, well-organised explanation of the method and why it works, and explains why trigonometry is important with two or more Kenyan examples.	Explains the method and gives one example of why trigonometry is important.	The explanation is incomplete or has no real-life examples.

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